Milestone 3

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Model Evaluation and Interpretation

Model Evaluation

For Milestone 2, I trained 3 machine learning models to predict video games based on song input using training data. The training data was a preprocessed set containing song and video game matchings based on the emotional scores and engagement scores I calculated previously. I briefly evaluated the models performance when matching the test data based on the Sklearn metrics built in accuracy, F1, and precision scores. Here is what those evaluation scores looked like:

Model	Accuracy	F1- Score	Precision
Random Forest	99.85%	0.9982	0.9980
Linear SVM	75.04%	0.0235	0.0155
Neural Network	20.30%	0.1259	0.1219

Figure 1: Model Evaluation with Accuracy, F1, and precision scores

As we can see, the Random Forest model outperformed Linear SVM and Neural network by a significantly higher percentage. This is most likely due to the ensemble structure of random forest that was highly capable of managing the complexity of the given task. The Linear SVM, in this scenario, acted as a baseline for average model performance. However, Neural Networks incredibly underperformed. This is a reasonable performance expectation, because neural networks are not as capable of handling dimensional data such as the dataset used.

Model Interpretation

Each of the models demonstrated different behaviors when mapping emotional and engagement scores from songs to video games. The Random Forest model consistently performed the best across the three metrics. The ensemble method nature of Random Forest was sufficient in evaluating the complex relationships between features. For that reason, Random Forest is the best model to use for this project. On the other hand, linear SVM scored relatively low on F1 and precision, despite showing moderately high accuracy. This suggests that the model might have defaulted to predicting the majority class, scoring decent accuracy due to class imbalance but failing to meaningfully capture minority class distinctions. Compared to the Random Forest and SVM, the Neural Network performed worse in accuracy, but was more precise than the linear SVM. Overall, this model performed the worst. I believe this is due to the fact that neural networks are sensitive to data dimensionality. So, mapping of emotional scores and engagement scores was most likely too high dimensional for the neural network to handle. Overall, the models indicated that emotional and engagement scores serve as effective features for genre mapping. However, it is important to consider class imbalances and model selection to properly evaluate the nature of subjective fields - such as personalized recommendations.

Biases and Limitations

The data used to train the models was somewhat limited - I was unable to match every single song in existence with every single video game in existence. So the models are biased and limited to what the dataset offers. Firstly, I'd like to examine some limitations with the given dataset. I already mentioned that not every song and video game that exists can be evaluated by the models. This also means that not every genre of song/video game is listed in the dataset, so the predominantly available data covers mainstream music and video games. This means the data is limited to popular songs and video games, and that's not always the answer. If a person who liked more unique video games/music used the dashboard - they may be disappointed by the results. Ideally, if I made this project to a larger scale I would have incorporated the google search function. I was not able to get that working by the due date, but in a perfect world the model would examine previously matched data, and search the internet for information to calculate the emotional and engagement scores, and match it to a video game.

These limitations are directly related to the biases of the model. For example, within the Hidden Trends in Video Games dataset, the most commonly occurring game genre is Sports at 16%. The next most common genre is Action at 14%. This essentially means there are more sporty video games within the dataset than action, and any other video game genres. The imbalance in data means the model may favor sports recommendations over less-represented genres - like puzzle, strategy, or rpg games. These data imbalances exist for each column in each dataset, no data is 100% equally represented. This is mirroring of real life data, where two categories of data are rarely represented equally.

Tool Development

For this project, I believed a dashboard would be the best outlet to display the findings and interact with the data. I wanted to get creative and display the cross domain project in a fun way. I used streamlit for python to create an interactive dashboard. The UI design is inspired by many video games and album covers conglomerating into one aesthetically pleasing - if not excessive - interface. Many of my favorite video games and album covers are purple or incorporate purple into their main color scheme. The dashboard features several purple orbs that float around and elastically collide with each other and the sides of the screen. The main purple orb in the center has a text box, prompting the user to input a song. The dashboard will then show a video game that the machine learning algorithm selects for the user. If the user selects a song that is not in the database, a song not found message will pop up, and prompt the user to try a different song. The surrounding orbs will showcase different songs, giving the user an idea of what kind of songs are available from the dataset. When a song is successfully chosen and matched, the match will appear on the orb with a surrounding orb displaying accuracy metrics.

References:

1. Sci-kit Learn documentation: http://scikit-learn.org/stable/index.html
2. Medium Articles:
    a. https://medium.com/geekculture/random-forest-ensemble-method-860aaf4fcd16
    b. http://medium.com/@anurag-jain/tf-idf-vectorization-with-cosine-similarity-eca3386d4423
    c. https://medium.com/data-science/how-to-bin-numerical-data-with-pandas-fe5146c9dc55

d. https://viniciusarz.medium.com/mastering-correlation-analysis-for-data-science-pearson-vs-spearman-cb1cc0a7b456
e. https://medium.com/data-science/feature-selection-with-pandas-e3690ad8504b

LLM Declarations:

I am not the best with frontend/visual aesthetics using code. I used chatGPT to help me create a visually aesthetic dashboard to meet my lofty aspirations for the project. Here are the prompts I used to assist me in the process:

“If I want to create a dashboard using streamlit that displays one main purple circle, 5 surrounding smaller circles, and 3 even smaller circles, how can I go about this? Describe the process to me without showing too much code.”

AI response: “Step-by-Step Plan to Create Your Streamlit Dashboard
1. Set up your Streamlit environment
2. Design the Circle Layout
3. Make It Interactive
4. Handle Circle Movement and Collisions (Optional)”, with information on how to achieve those tasks. I used this step by step process to create the dashboard on my own.

I also wanted to find a good way to explain why the linear SVM model was moderately accurate, yet super low scoring on F1 and Precision metrics.

Prompt: Why would a linear SVM model perform somewhat accurately, yet score low on F1 and Precision metrics?

AI Response:
“The model is basically **guessing the most common class label** for everything — because it either:

- Can’t learn enough from the data
- Is overwhelmed by **class imbalance**
- Or is underfit and defaulting to the "safest" guess”

I then researched more on predicting the majority class, and found that this is most likely what the model is doing.